Digital Electronics Question and Answers

Registers

Topic 9

1.) Registers

2.) Shift Registers

3.) Universal Shift Registers

4.) Shift Register Counters

5.) Ring Counter

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Digital Electronics Question & Answers

1.) Registers

1. A register is defined as ___________
a) The group of latches for storing one bit of information
b) The group of latches for storing n-bit of information
c) The group of flip-flops suitable for storing one bit of information
d) The group of flip-flops suitable for storing binary information
Answer: d
Explanation: A register is defined as the group of flip-flops suitable for storing binary information. Each flip-flop is a binary cell capable of storing one bit of information. The data in a register can be transferred from one flip-flop to another.

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2. The register is a type of ___________
a) Sequential circuit
b) Combinational circuit
c) CPU
d) Latches
Answer: a
Explanation: Register's output depends on the past and present states of the inputs. The device which follows these properties is termed as a sequential circuit. Whereas, combinational circuits only depend on the present values of inputs.

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3. How many types of registers are?
a) 2
b) 3
c) 4
d) 5

Answer: c
Explanation: There are 4 types of shift registers, viz., Serial-In/Serial-Out, Serial-In/Parallel-Out, Parallel-In/Serial-Out and Parallel-In/Parallel-Out.

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4. The main difference between a register and a counter is ___________
a) A register has no specific sequence of states
b) A counter has no specific sequence of states
c) A register has capability to store one bit of information but counter has n-bit
d) A register counts data
Answer: a

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Explanation: The main difference between a register and a counter is that a register has no specific sequence of states except in certain specialised applications.

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5. In D register, 'D' stands for ____________
a) Delay
b) Decrement
c) Data
d) Decay
Answer: c
Explanation: D stands for "data" in case of flip-flops and not delay. Registers are made of a group of flip-flops.

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6. Registers capable of shifting in one direction is ____________
a) Universal shift register
b) Unidirectional shift register
c) Unipolar shift register
d) Unique shift register

Answer: b
Explanation: The register capable of shifting in one direction is unidirectional shift register. The register capable of shifting in both directions is known as a bidirectional shift register.

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7. A register that is used to store binary information is called ____________
a) Data register
b) Binary register
c) Shift register
d) D – Register
Answer: b
Explanation: A register that is used to store binary information is called a binary register. A register in which data can be shifted is called shift register.

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8. A shift register is defined as ____________
a) The register capable of shifting information to another register
b) The register capable of shifting information either to the right or to the left
c) The register capable of shifting information to the right only
d) The register capable of shifting information to the left only

Answer: b
Explanation: The register capable of shifting information either to the right or to the left is termed as shift register. A register in which data can be shifted only in one direction is

called unidirectional shift register, while if data can shifted in both directions, it is known as a bidirectional shift register.

9. How many methods of shifting of data are available?
a) 2
b) 3
c) 4
d) 5
Answer: a
Explanation: There are two types of shifting of data are available and these are serial shifting & parallel shifting.

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10. In serial shifting method, data shifting occurs ____________
a) One bit at a time
b) simultaneously
c) Two bit at a time
d) Four bit at a time

Answer: a
Explanation: As the name suggests serial shifting, it means that data shifting will take place one bit at a time for each clock pulse in a serial fashion. While in parallel shifting, shifting will take place with all bits simultaneously for each clock pulse in a parallel fashion.

2.) Shift Registers

1. Based on how binary information is entered or shifted out, shift registers are classified into _______ categories.
a) 2
b) 3
c) 4
d) 5
Answer: c
Explanation: The registers in which data can be shifted serially or parallelly are known as shift registers. Based on how binary information is entered or shifted out, shift registers are classified into 4 categories, viz., Serial-In/Serial-Out(SISO), Serial-In/Parallel-Out (SIPO), Parallel-In/Serial-Out (PISO), Parallel-In/Parallel-Out (PIPO).

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2. The full form of SIPO is ___________
a) Serial-in Parallel-out
b) Parallel-in Serial-out

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c) Serial-in Serial-out
d) Serial-In Peripheral-Out

Answer: a
Explanation: SIPO is always known as Serial-in Parallel-out.

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3. A shift register that will accept a parallel input or a bidirectional serial load and internal shift features is called as?
a) Tristate
b) End around
c) Universal
d) Conversion

Answer: c
Explanation: A shift register can shift it's data either left or right. The universal shift register is capable of shifting data left, right and parallel load capabilities.

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4. How can parallel data be taken out of a shift register simultaneously?
a) Use the Q output of the first FF
b) Use the Q output of the last FF
c) Tie all of the Q outputs together
d) Use the Q output of each FF

Answer: d
Explanation: Because no other flip-flops are connected with the output Q, therefore one can use the Q out of each FF to take out parallel data.

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5. What is meant by the parallel load of a shift register?
a) All FFs are preset with data
b) Each FF is loaded with data, one at a time
c) Parallel shifting of data
d) All FFs are set with data

Answer: a
Explanation: At Preset condition, outputs of flip-flops will be 1. Preset = 1 means Q = 1, thus input is definitely 1.

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6. The group of bits 11001 is serially shifted (right-most bit first) into a 5-bit parallel output shift register with an initial state 01110. After three clock pulses, the register contains

____
a) 01110
b) 00001

c) 00101
d) 00110

Answer: c
Explanation: LSB bit is inverted and feed back to MSB:
01110->initial
10111->first clock pulse
01011->second
00101->third.

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7. Assume that a 4-bit serial in/serial out shift register is initially clear. We wish to store the nibble 1100. What will be the 4-bit pattern after the second clock pulse? (Right-most bit first)
a) 1100
b) 0011
c) 0000
d) 1111

Answer: c
Explanation: In Serial-In/Serial-Out shift register, data will be shifted one at a time with every clock pulse. Therefore,
Wait | Store
1100 | 0000
110 | 0000 1st clock
11 | 0000 2nd clock.

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8. A serial in/parallel out, 4-bit shift register initially contains all 1s. The data nibble 0111 is waiting to enter. After four clock pulses, the register contains ________
a) 0000
b) 1111
c) 0111
d) 1000

Answer: c
Explanation: In Serial-In/Parallel-Out shift register, data will be shifted all at a time with every clock pulse. Therefore,
Wait | Store
0111 | 0000
011 | 1000 1st clk
01 | 1100 2nd clk

0 | 1110 3rd clk

X | 1111 4th clk.

---

9. With a 200 kHz clock frequency, eight bits can be serially entered into a shift register in

____

a) 4 µs

b) 40 µs

c) 400 µs

d) 40 ms

Answer: b

Explanation: f = 200 KHZ; T = (1/200) m sec = (1/0.2) micro-sec = 5 micro-sec;

In serial transmission, data enters one bit at a time. After 8 clock cycles only 8 bit will be loaded = 8 * 5 = 40 micro-sec.

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10. An 8-bit serial in/serial out shift register is used with a clock frequency of 2 MHz to achieve a time delay (td) of ________

a) 16 us

b) 8 us

c) 4 us

d) 2 us

Answer: c

Explanation: One clock period is = (½) micro-s = 0.5 microseconds. In serial transmission, data enters one bit at a time. So, the total delay = 0.5*8 = 4 micro seconds time is required to transmit information of 8 bits.

3.) Universal Shift Registers

1. A sequence of equally spaced timing pulses may be easily generated by which type of counter circuit?

a) Ring shift

b) Clock

c) Johnson

d) Binary

Answer: a

Explanation: In Ring counter, the feedback of the output of the FF is fed to the same FF's input. Thus, it generates equally spaced timing pulses.

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2. A bidirectional 4-bit shift register is storing the nibble 1101. Its input is HIGH. The nibble 1011 is waiting to be entered on the serial data-input line. After three clock pulses, the shift register is storing ________

a) 1101
b) 0111
c) 0001
d) 1110

Answer: b
Explanation: Mode is high means it's a right shift register. Then after 3 clock pulses enter bits are 011 and remained bit in register is 1. Therefore, 0111 is the required solution.
1011 | 1101
101 | 1110 -> 1st clock pulse
10 | 1111 -> 2nd clock pulse
1 | 0111 -> 3rd clock pulse.

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3. To operate correctly, starting a ring shift counter requires ___________
a) Clearing all the flip-flops
b) Presetting one flip-flop and clearing all others
c) Clearing one flip-flop and presetting all others
d) Presetting all the flip-flops
Answer: b
Explanation: In Ring counter, the feedback of the output of the FF is fed to the same FF's input. To operate correctly, starting a ring shift counter requires presetting one flip-flop and clearing all others so that it can shift to the next bit.

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4. A 4-bit shift register that receives 4 bits of parallel data will shift to the _________ by _________ position for each clock pulse.
a) Right, one
b) Right, two
c) Left, one
d) Left, three

Answer: a
Explanation: If register shifts towards left then it shift by a bit to the left and if register shifts right then it shift to the right by one bit. Since, it receives parallel data, then by default, it will shift to right by one position.

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5. How many clock pulses will be required to completely load serially a 5-bit shift register?
a) 2
b) 3
c) 4
d) 5
Answer: d

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Explanation: A register is a collection of FFS. To load a bit, we require 1 clock pulse for 1 shift register. So, for 5-bit shift register we would require of 5 clock pulses.

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6. How is a strobe signal used when serially loading a shift register?
a) To turn the register on and off
b) To control the number of clocks
c) To determine which output Qs are used
d) To determine the FFs that will be used
Answer: b
Explanation: A strobe is used to validate the availability of data on the data line. It (an auxiliary signal used to help synchronize the real data in an electrical bus when the bus components have no common clock) signal is used to control the number of clocks during serially loading a shift register.

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7. An 8-bit serial in/serial out shift register is used with a clock frequency of 150 kHz. What is the time delay between the serial input and the Q3 output?
a) 1.67 s
b) 26.67 s
c) 26.7 ms
d) 267 ms
Answer: c
Explanation: In serial-shifting, one bit of data is shifted one at a time. From Q0 to Q3 total of 4 bit shifting takes place. Therefore, 4/150kHz = 26.67 microseconds.

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8. What are the three output conditions of a three-state buffer?
a) HIGH, LOW, float
b) High-Z, 0, float
c) Negative, positive, 0
d) 1, Low-Z, float

Answer: a
Explanation: Three conditions of a three-state buffer are HIGH, LOW & float.

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9. The primary purpose of a three-state buffer is usually _____________
a) To provide isolation between the input device and the data bus
b) To provide the sink or source current required by any device connected to its output without loading down the output device
c) Temporary data storage
d) To control data flow

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Answer: a

Explanation: The primary purpose of a three-state buffer is usually to provide isolation between the input device or peripheral devices and the data bus. Three conditions of a three-state buffer are HIGH, LOW & float.

10. What is the difference between a ring shift counter and a Johnson shift counter?
a) There is no difference
b) A ring is faster
c) The feedback is reversed
d) The Johnson is faster

Answer: c

Explanation: A ring counter is a shift register (a cascade connection of flip-flops) with the output of the last one connected to the input of the first, that is, in a ring. Whereas, a Johnson counter (or switchtail ring counter, twisted-ring counter, walking-ring counter, or Moebius counter) is a modified ring counter, where the output from the last stage is inverted and fed back as input to the first stage.

4.) Shift Register Counters

1. What is a recirculating register?
a) Serial out connected to serial in
b) All Q outputs connected together
c) A register that can be used over again
d) Parallel out connected to Parallel in

Answer: a

Explanation: A recirculating register is a register whose serial output is connected to the serial input in a circulated manner.

2. When is it important to use a three-state buffer?
a) When two or more outputs are connected to the same input
b) When all outputs are normally HIGH
c) When all outputs are normally LOW
d) When two or more outputs are connected to two or more inputs

Answer: a

Explanation: When two or more outputs are connected to the same input, in such situation we use of tristate buffer always because it has the capability to take upto three inputs. A buffer is a circuit where the output follows the input.

3. A bidirectional 4-bit shift register is storing the nibble 1110. Its input is LOW. The nibble 0111 is waiting to be entered on the serial data-input line. After two clock pulses, the shift

register is storing ________
a) 1110
b) 0111
c) 1011
d) 1001

Answer: c
Explanation: Given,

Waiting nibble | Stored nibble

0111 | 1110, Initially

011 | 1101, 1st pulse (left shift)

01 | 1011, 2nd pulse (left shift)

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4. In a parallel in/parallel out shift register, D0 = 1, D1 = 1, D2 = 1, and D3 = 0. After three clock pulses, the data outputs are ________
a) 1110
b) 0001
c) 1100
d) 1000
Answer: b
Explanation: Parallel in parallel out gives the same output as input. Thus, after three clock pulses, the data outputs are 0001.

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5. The group of bits 10110111 is serially shifted (right-most bit first) into an 8-bit parallel output shift register with an initial state 11110000. After two clock pulses, the register contains ______________
a) 10111000
b) 10110111
c) 11110000
d) 11111100
Answer: d
Explanation: After first clock pulse, the register contains 11111000. After second clock pulse, the register would contain 11111100. Since the bits are shifted to the right at every clock pulse.

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6. By adding recirculating lines to a 4-bit parallel-in serial-out shift register, it becomes a ________ ________ and ________ out register.
a) Parallel-in, serial, parallel
b) Serial-in, parallel, serial

c) Series-parallel-in, series, parallel
d) Bidirectional in, parallel, series
Answer: a
Explanation: One bit shifting takes place just after the output obtained on every register. Hence, by adding recirculating lines to a 4-bit parallel-in serial-out shift register, it becomes a Parallel-in, Serial, and Parallel-out register. Since, the bots can be inputted all at the same time, while the data can be outputted either one at a time or simultaneously.

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7. What type of register would have a complete binary number shifted in one bit at a time and have all the stored bits shifted out one at a time?
a) Parallel-in Parallel-out
b) Parallel-in Serial-out
c) Serial-in Serial-out
d) Serial-in Parallel-out
Answer: c
Explanation: Serial-in Serial-out register would have a complete binary number shifted in one bit at a time and have all the stored bits shifted out one at a time. Since in serial transmission, bits are transmitted or received one at a time and not simultaneously.

---

8. In a 4-bit Johnson counter sequence, there are a total of how many states or bit patterns?
a) 1
b) 3
c) 4
d) 8

Answer: d
Explanation: In johnson counter, total number of states are determined by 2^N = 2*4 = 16
Total Number of Used states = 2N = 2*4 = 8
Total Number of Unused states = 16 – 8 = 8.

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9. If a 10-bit ring counter has an initial state 1101000000, what is the state after the second clock pulse?
a) 1101000000
b) 0011010000
c) 1100000000
d) 0000000000

Answer: b
Explanation: After shifting 2-bit we get the output as 0011010000 (Since two zeros are at 1^{st} position and 2^{nd} position which came from the last two bits). As in a ring counter, the bits rotate in clockwise direction.

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10. How much storage capacity does each stage in a shift register represent?
a) One bit
b) Two bits
c) Four bits
d) Eight bits

Answer: a
Explanation: A register is made of flip-flops. And each flip-flop stores 1 bit of data. Thus, a shift register has the capability to store one bit and if another bit is to store, in such a situation it deletes the previous data and stores them.

5.) Ring Counter

1. Ring shift and Johnson counters are ____________
a) Synchronous counters
b) Asynchronous counters
c) True binary counters
d) Synchronous and true binary counters

Answer: a
Explanation: Synchronous counters are the counters being triggered in the presence of a clock pulse. Since all of the clock inputs are connected through a single clock pulse in ring shift and johnson counters. So, both are synchronous counters.

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2. What is the difference between a shift-right register and a shift-left register?
a) There is no difference
b) The direction of the shift
c) Propagation delay
d) The clock input
Answer: b
Explanation: In shift-right register, shifting of bit takes place towards the right and towards left for shift-left register. Thus, both the registers vary in the shifting of their direction.

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3. What is a transceiver circuit?
a) A buffer that transfers data from input to output
b) A buffer that transfers data from output to input
c) A buffer that can operate in both directions
d) A buffer that can operate in one direction

Answer: c

Explanation: A transceiver circuit is a buffer that can operate in both directions right as well as left.

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4. A 74HC195 4-bit parallel access shift register can be used for ____________

a) Serial in/serial out operation

b) Serial in/parallel out operation

c) Parallel in/serial out operation

d) All of the Mentioned

Answer: d

Explanation: 74HC195 is an IC, which can be used for all of the given operations, as well as for, parallel-in/parallel-out.

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5. Which type of device may be used to interface a parallel data format with external equipment's serial format?

a) UART

b) Key matrix

c) Memory chip

d) Series in Parallel out

Answer: a

Explanation: UART means Universal Asynchronous Receiver/Transmitter which converts the bytes it receives from the computer along parallel circuits into a single serial bit stream for outbound transmission. And also receives data in serial form and converts it into parallel form and sent to the processor.

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6. What is the function of a buffer circuit?

a) To provide an output that is inverted from that on the input

b) To provide an output that is equal to its input

c) To clean up the input

d) To clean up the output

Answer: b

Explanation: The function of a buffer circuit is to provide an output that is equal to its input. A transceiver circuit is a buffer that can operate in both directions right as well as left.

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7. What is the preset condition for a ring shift counter?

a) All FFs set to 1

b) All FFs cleared to 0

c) A single 0, the rest 1

d) A single 1, the rest 0

Answer: d

Explanation: A ring shift counter is a counter in which the output of one FF connected to the input of the adjacent FF. In preset condition, all of the bits are 0 except first one.

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8. Which is not characteristic of a shift register?
a) Serial in/parallel in
b) Serial in/parallel out
c) Parallel in/serial out
d) Parallel in/parallel out

Answer: a

Explanation: There is no such type of register present who doesn't have output end. Thus, Serial in/Parallel in is not a characteristic of a shift register. There has to be an output, be it serial or parallel.

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9. To keep output data accurate, 4-bit series-in, parallel-out shift registers employ a ____________
a) Divide-by-4 clock pulse
b) Sequence generator
c) Strobe line
d) Multiplexer

Answer: c

Explanation: In computer or memory technology, a strobe is a signal that is sent that validates data or other signals on adjacent parallel lines. Thus, in registers the strobe line is there to check the availability of data.

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10. Another way to connect devices to a shared data bus is to use a ____________
a) Circulating gate
b) Transceiver
c) Bidirectional encoder
d) Strobed latch

Answer: b

Explanation: A transceiver is a device comprising both a transmitter and a receiver which are combined and share common circuitry or a single housing. When no circuitry is common between transmit and receive functions, the device is a transmitter-receiver.
